A Tiny Robot That Swims With Living Muscles

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Imagine a robot powered not by batteries and motors, but by living muscle. Scientists at MIT have built exactly that: a tiny swimming robot that flaps its fins using a layer of real muscle cells.

The robot's body is a flexible gel frame about the size of a stick of gum. It has two fins, and each fin is coated with muscle cells far thinner than a human hair. The cells were **genetically modified** so that they squeeze tight when light shines on them. By flashing light onto one fin or the other, and changing how often the light pulses, the researchers can steer the robot and control its speed. In tests, the paper-thin robot swam and turned through a simple underwater maze. At top speed it covered about four times its own body length every minute. An Olympic swimmer manages up to 65 body lengths per minute, and the little robot's pace is similar to a cow shark cruising lazily through the ocean.

"It takes a lot of force to move through water versus air," said study author Ritu Raman, an **associate professor** of **mechanical engineering** at MIT. "The robot's quite strong, given its size." The study, published in the journal Advanced Functional Materials, is the first demonstration of a very thin, flat robot that can travel from place to place using living muscle tissue. Most earlier **biohybrid** robots were built from bulky chunks of **lab-grown** muscle needing millions of cells. The MIT team believes thinner designs could be cheaper to build and move more efficiently. One day, Raman says, robots powered by living muscle might do delicate jobs in fragile places where ordinary machines are too rough, because living tissue is soft, responds to its surroundings, and can even heal itself.

Word Treasure

genetically modified -- Changed by scientists altering the instructions inside living cells.

biohybrid -- A machine that mixes living tissue with built, non-living parts.

lab-grown -- Made or grown in a laboratory instead of inside a living body.

mechanical engineering -- The branch of engineering that designs and builds machines and their moving parts.

associate professor -- A university teacher and researcher ranked below a full professor.

Background you need

Biohybrid robots are machines that combine living cells with built materials, so real tissue does part of the work instead of a motor.

Muscle moves by squeezing tight, so even a thin layer of muscle cells can flap a soft fin and push a small robot through water.

Scientists can change muscle cells so they react to light, and flashing light on one side or the other becomes a simple way to steer.

Because living tissue is soft, senses its surroundings and can heal itself, such robots could one day handle delicate jobs in fragile places where ordinary machines are too rough.

Break it down

LEAD: MIT scientists built a tiny swimming robot powered by living muscle cells instead of batteries and motors

KEY EVENT: In tests, the paper-thin robot swam and turned through a simple underwater maze

+- WHO: Study author Ritu Raman and her MIT team

+- HOW: Light flashes make genetically modified muscle cells on the fins squeeze tight, steering the robot and setting its speed

+- HOW FAST: About four body lengths per minute at top speed, similar to a cow shark cruising lazily

EVIDENCE: The study was published in the journal Advanced Functional Materials

OPEN QUESTION: Whether muscle-powered robots can one day do delicate jobs in fragile places where ordinary machines are too rough

Why it matters

Robots driven by living tissue could work gently in fragile places that hard machines might damage, opening a new way to explore and help in delicate environments. The work also suggests that thinner, simpler designs may make such biohybrid robots cheaper to build and more efficient.

Test what you remember

1. What powers the tiny robot's swimming?

- (A) Living muscle cells on its fins
- (B) A spinning propeller
- (C) Magnets under the water
- (D) A tiny battery and motor

2. How big is the robot's gel frame?

- (A) About the size of a stick of gum
- (B) As big as a shoe
- (C) The size of a small boat
- (D) As long as a human arm

3. How do the researchers steer the robot?

- (A) By flashing light onto one fin or the other and changing how often it pulses
- (B) By shouting voice commands
- (C) By pulling it with a string
- (D) By changing the water temperature

4. How fast did the robot swim at top speed?

- (A) 65 body lengths per minute
- (B) One body length per hour
- (C) Faster than a cow shark
- (D) About four body lengths per minute

5. What did the robot do during the tests?

- (A) It raced an Olympic swimmer
- (B) It swam and turned through a simple underwater maze
- (C) It jumped out of the water
- (D) It collected plastic waste

6. What makes this study a first, according to the article?

- (A) It is the first robot built at MIT
- (B) It is the first robot ever to swim
- (C) It is the first very thin, flat robot that can travel from place to place using living muscle tissue
- (D) It is the first robot that can heal itself

Answer key (try the quiz first!): 1: A 2: A 3: A 4: D 5: B 6: C

Questions to think about

1. Ritu Raman (study author): Says the robot is quite strong for its size and that muscle-powered robots might one day do delicate jobs in fragile places because living tissue is soft, responsive and can heal itself.

2. The MIT team: Believes thinner designs could be cheaper to build and move more efficiently than earlier biohybrid robots made from bulky chunks of lab-grown muscle.

MY THOUGHTS (at least 20 words):